

Anjali Deshmukh

AI Engineer | Agent Systems & Production ML

Portland, OR | (503) 441-2876 | anjali.deshmukh@mailbox.com

EXPERIENCE

Senior AI Engineer | Vortex Commerce — Portland, OR | Mar 2024 – Aug 2026

- Designed and deployed autonomous order management agent serving 12,000+ daily transactions across 8 retail partners; agent uses tool calling to query inventory APIs, trigger fulfillment workflows, and escalate edge cases to human operators
- Built memory layer for customer context persistence, enabling agent to handle multi-turn conversations about order status, returns, and shipping preferences without repeating information
- On-call for production incidents; implemented monitoring dashboards tracking agent success rate (now 94%), latency, and tool invocation patterns; reduced customer escalations by 31% through iterative prompt refinement and tool definition improvements
- Presented agent architecture and performance metrics to C-suite and board advisors; demonstrated live system handling real customer inquiries during quarterly reviews
- Integrated Shopify, WooCommerce, and custom POS APIs into agent orchestration layer; handled authentication, rate limiting, and retry logic across heterogeneous systems

ML Engineer | Meridian Retail Labs — Seattle, WA | Jul 2022 – Feb 2024

- Shipped personalization recommendation engine live to 45 e-commerce sites (8M+ monthly active users); built using multi-armed bandit approach with daily retraining on user interaction signals
- Owned monitoring and incident response for recommendation pipeline; implemented alerts on model drift, API latency, and serving infrastructure; debugged and patched cold-start recommendation issues affecting 2% of new users
- Collaborated with product and merchant teams to define recommendation objectives; conducted monthly stakeholder reviews translating model performance metrics into business KPIs (AOV, conversion lift)
- Integrated with Elasticsearch and Redis for real-time feature lookups and candidate ranking; optimized P99 latency from 650ms to 180ms through caching strategy refinement

AI/ML Intern → Software Engineer | Cascadia Data Systems — Tacoma, WA | Jun 2021 – Jun 2022

- Developed data pipelines and feature engineering for customer churn prediction model used by 6 B2B SaaS clients; validated model performance on hold-out sets and documented decision thresholds for deployment
- Built API endpoints for model inference; designed graceful degradation when upstream services failed
- Shadowed and supported customer success team on client calls; translated technical constraints and model capabilities into language business teams understood

EDUCATION

B.S. Computer Science | University of Washington | 2021

GPA: 3.7 | Relevant coursework: Machine Learning, Distributed Systems, Software Engineering

TECHNICAL SKILLS

Python, PyTorch, LangChain, LLMs, REST APIs, PostgreSQL, AWS (Lambda, SageMaker, RDS), Kubernetes, Git, SQL, feature engineering, agent orchestration, prompt engineering